Vehicle and Passenger Safety from Confined Water Logging

Team Name: Nexora | **Team Members:** Abinaya S, Indhu S, Anees Fathima B I | **Department:** Electronics and Communication Engineering | **Institution:** Bannari Amman Institute of Technology, Sathyamangalam.

Problem Statement

Urban waterlogging creates severe safety risks in underpasses and low-lying roads during monsoon seasons and extreme weather events. Sudden water accumulation traps vehicles and passengers in life-threatening conditions.

Sudden Water Accumulation

Rapid flooding during heavy rainfall overwhelms drainage systems in urban areas

Vehicle Stalling

Motorists entering flooded roads unaware of depth may experience engine failure

Passenger Entrapment

Occupants become trapped inside submerged vehicles in dangerous conditions

Monitoring Gap

Absence of continuous real-time monitoring systems for flood-prone zones

Delayed Response

Emergency services receive late notifications, delaying rescue operations

Need for the Project

Urban underpasses and low-lying roads frequently experience sudden waterlogging during heavy rainfall. A real-time monitoring and automated detection system is required to identify dangerous flood conditions and prevent vehicles from entering flooded areas.

Climate Crisis

Increasing extreme rainfall events and inadequate urban drainage systems lead to frequent waterlogging in city roads and underpasses.

Infrastructure Modernization

Smart city requirements demand automated monitoring and intelligent response systems

Real-Time Detection

Continuous flood level monitoring improves road safety and prevents accidents

Rapid Emergency Response

Automated alerts reduce rescue response time and save lives

Damage Prevention

Early warnings prevent vehicle damage and minimize traffic disruption

Project Objectives

01

Detect Waterlogging Levels

Measure real-time water depth in flood-prone underpasses using an **HC-SR04 ultrasonic sensor** connected to a **Raspberry Pi**, and classify conditions into **safe, warning, and critical levels**.

02

Continuous Underpass Monitoring

Deploy a **Raspberry Pi–based monitoring system** with an integrated camera module to continuously observe flood-prone road sections and capture visual data during rising water levels.

03

Identify Trapped Vehicles

Analyze captured image frames using **YOLO-based deep learning object detection** to identify vehicles present within flooded areas and determine potential risk conditions.

04

Deep Learning Detection

Implement **YOLO (You Only Look Once) models optimized with Particle Swarm Optimization (PSO)** to improve detection accuracy and real-time performance on the embedded platform.

05

Generate Emergency Alerts

When critical water levels and vehicle presence are detected, automatically send **GPS-based emergency alerts via GSM and Telegram notification systems** to authorities and nearby users.

System Methodology

Environmental Sensing, Image Processing, and Deep Learning–Based Object Detection

The proposed system follows a structured methodology that integrates **environmental sensing, image processing, and deep learning–based object detection** to monitor waterlogging conditions in urban road environments. The system continuously collects environmental data from sensing units installed in flood-prone locations and analyzes the information to determine potential flood risks.

Edge Computing Platform

Sensor readings and visual inputs are processed through an **edge computing platform**, enabling real-time monitoring without relying on centralized cloud processing. Image frames captured from the monitoring camera are analyzed using deep learning techniques to detect vehicles present in flooded regions.

Combined Sensor Measurements and Visual Detection

The system combines **sensor measurements and visual detection results** to assess whether vehicles are at risk due to rising water levels. Based on this combined analysis, the system can automatically identify critical situations and trigger appropriate response mechanisms. This methodology enables **continuous monitoring, faster decision-making, and improved safety for vehicles and passengers in flood-prone areas**.

System Implementation

Step 1: Water Level Detection

An HC-SR04 ultrasonic sensor, installed inside a sensing pipe, measures the distance to the water surface by emitting ultrasonic pulses. The Raspberry Pi calculates the time taken for the signal to return, converting this data into real-time water depth measurements to monitor flood conditions.

Step 2: Data Processing

The Raspberry Pi 4 Model B serves as the central processing unit, continuously receiving sensor data. It compares these measurements against predefined thresholds to classify the flooding situation into safe, warning, or critical conditions.

Step 3: Camera Activation

Once the water level reaches the critical threshold, the system automatically activates the Raspberry Pi camera module positioned above the road. This captures real-time visual frames of the flooded section for immediate traffic analysis.

Step 4: Vehicle Detection

The system processes captured image frames using YOLO deep learning models implemented in PyTorch. The algorithm identifies vehicles within the flooded region by calculating bounding boxes and classification probabilities in the visual data.

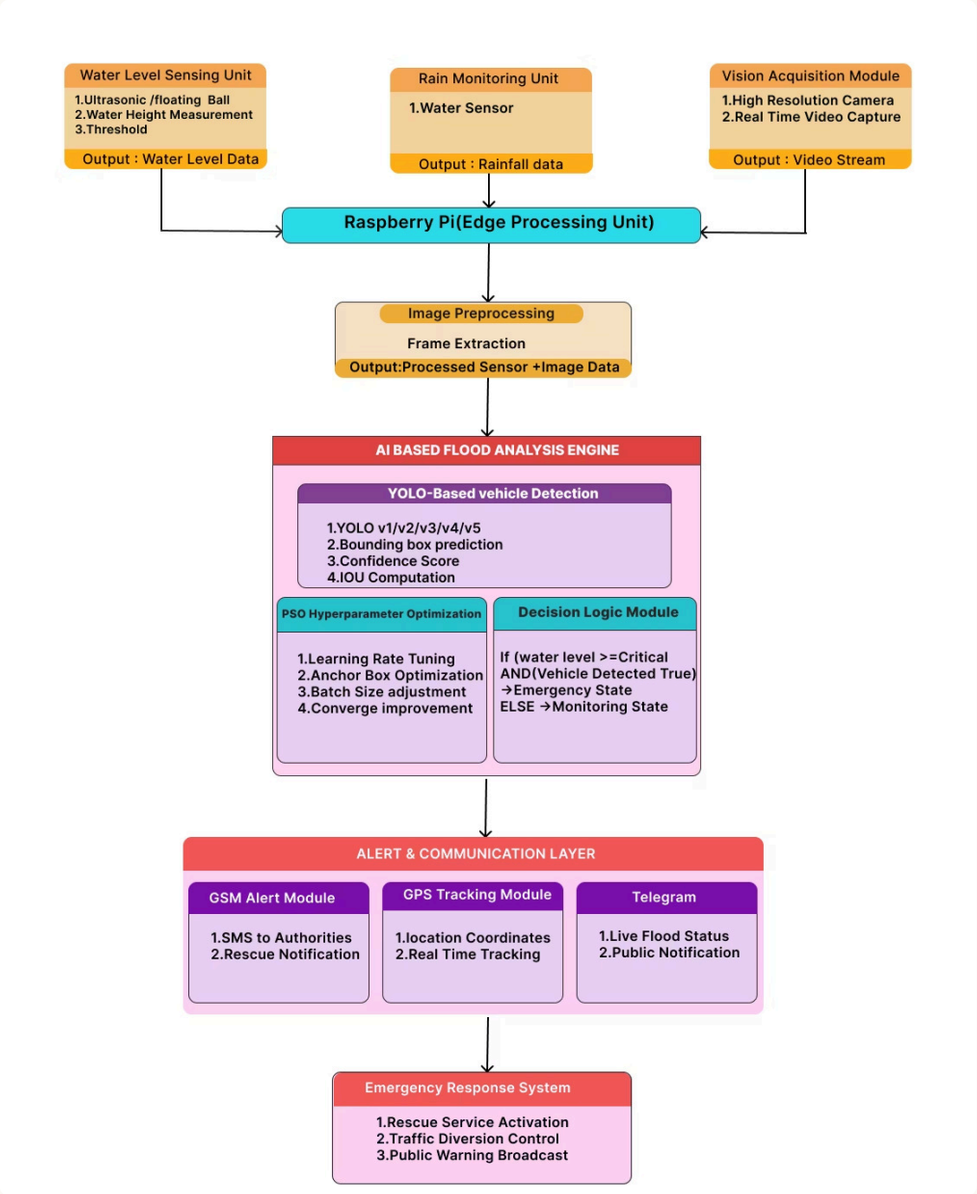
Step 5: Risk Identification

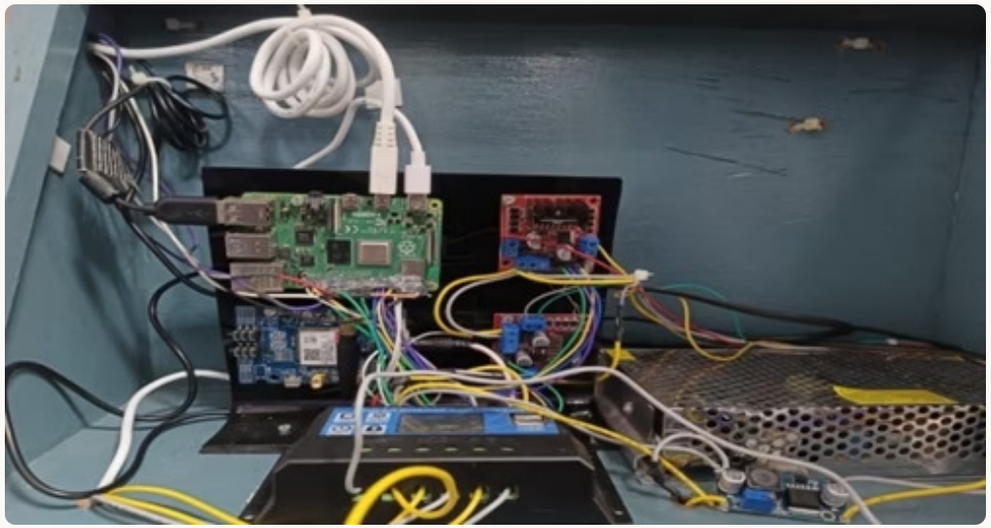
The system integrates sensor data with vehicle detection results. If vehicles are identified in areas where the water level exceeds safe thresholds, the system flags the situation as a high-priority safety risk for passengers.

Step 6: Emergency Alert System

Upon detecting a risk, the system retrieves precise location coordinates via GPS and triggers the GSM module. Notifications, including Telegram alerts and SMS messages, are sent to authorities to enable rapid rescue and traffic control.

Block Diagram and Hardware Setup





The system is centered around a **Raspberry Pi 4 Model B**, which interfaces with all sensing and communication modules through its GPIO and serial interfaces. The **HC-SR04 ultrasonic sensor** is connected to the Raspberry Pi using GPIO pins (TRIG and ECHO) to measure the real-time water level inside the sensing pipe. The **Raspberry Pi Camera Module** is connected through the **CSI (Camera Serial Interface)** port to capture video frames of the underpass for computer vision analysis. A **GPS module communicates via UART serial interface** to provide location coordinates, while the **GSM module transmits emergency alerts using AT commands over serial communication**. All sensor readings and image data are processed locally on the Raspberry Pi to enable real-time flood monitoring and alert generation.

Technologies and Components

Hardware Components

Raspberry Pi 4

Embedded processing platform

HC-SR04 Ultrasonic Sensor

Water level measurement

Raspberry Pi Camera Module

Visual monitoring

GPS Module

Location tracking

GSM Module

Wireless communication

LED Indicators

Status visualization

Software Stack

Python

Primary programming language

PyTorch

Deep learning framework

YOLO v1-v5

Object detection models

Particle Swarm Optimization

Model parameter tuning

Performance Analysis

The performance of different YOLO models was evaluated for vehicle detection in flooded road environments. The models were compared using standard object detection metrics including **Precision, Recall, F1-Score, Accuracy, and Frames Per Second (FPS)** to determine the most suitable model for real-time flood monitoring.

Model	Precision	Recall	F1 Score	Accuracy	FPS
YOLOv1	78.4%	74.6%	76.4%	75.8%	45
YOLOv2	84.7%	82.3%	83.5%	83.1%	40
YOLOv3	90.2%	88.5%	89.3%	89.7%	30
YOLOv4	92.6%	91.4%	92.0%	91.8%	36
YOLOv5	94.1%	93.2%	93.6%	93.4%	60

Result:
Among all evaluated models, **YOLOv5 achieved the highest precision, recall, and overall accuracy while maintaining the fastest detection speed**, making it the most suitable model for real-time vehicle detection in flooded road monitoring systems.

Novelty and Conclusion

IoT + Deep Learning Integration

The proposed system integrates **IoT-based environmental sensing with deep learning–based computer vision**, enabling simultaneous monitoring of water levels and vehicle presence in flood-prone urban roads.

Real-Time Flood Monitoring

The system performs **continuous real-time monitoring of underpass water levels using ultrasonic sensing and edge computing**, allowing early detection of dangerous flood conditions.

YOLO-Based Vehicle Detection

A **YOLO-based object detection framework** is implemented to accurately identify vehicles present in flooded areas, enabling reliable detection even under challenging environmental conditions.

PSO-Based Optimization

Particle Swarm Optimization (PSO) is applied to optimize model hyperparameters such as learning rate and anchor boxes, improving detection accuracy and system efficiency.

GPS Emergency Alert System

When critical flood conditions and trapped vehicles are detected, the system automatically sends **location-aware emergency alerts through GSM and Telegram communication modules** to authorities for rapid response.

Conclusion: The proposed system provides an intelligent framework for **real-time urban waterlogging detection and vehicle safety monitoring**. By combining IoT sensing, edge computing, and deep learning–based object detection, the system enables early risk identification and automated emergency alerts, significantly improving **road safety and disaster response during urban flooding events**.